

```
> m[1, 1, 1]
[1] 1
```

For the given example, the values in 'a' are represented as a double.

```
> typeof(a)
[1] "double"
```

The class of 'a' is a 'matrix' 'array'.

```
> class(a)
[1] "matrix" "array"
```

Matrix

A matrix is a vector in two dimensions. You can use the *matrix()* function to create a new matrix, as shown below:

```
> m <- matrix(data=c(1, 2, 3, 4, 5, 6, 7, 8, 9), nrow=3,
ncol=3)
> m
      [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
```

The 'm' matrix has a 'double' value as its type.

```
> typeof(m)
[1] "double"
```

You can of course retrieve the individual members in the matrix by specifying the indices.

```
> m[1]
[1] 1
```

You can retrieve the first row of the matrix similar to an array, as shown below:

```
> m[1,]
[1] 1 4 7
```

A range of indices can be supplied to the matrix to return the relevant elements:

```
> m[1:2, ]
      [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
```

You can also specify all the dimensions to get a specific

value present in a matrix cell:

```
> m[1,2]
[1] 4
```

The assignment operator can be used to update a specific value in the matrix, as shown below:

```
> m[1,2] <- 100
> m
      [,1] [,2] [,3]
[1,]    1  100    7
[2,]    2    5    8
[3,]    3    6    9
```

You can also specify the dimension names when creating a matrix, as illustrated below:

```
> m <- matrix(data=c(1, 2, 3, 4, 5, 6, 7, 8, 9), nrow=3,
ncol=3, dimnames=list(c("r1", "r2", "r3"), c("c1", "c2",
"c3")))
> m
      c1 c2 c3
r1    1  4  7
r2    2  5  8
r3    3  6  9
```

The specific row names of the matrix can be displayed, as shown below:

```
> row.names(m)
[1] "r1" "r2" "r3"
```

The various attributes of the matrix can be obtained using the *attributes()* function:

```
> attributes(m)
$dim
[1] 3 3

$dimnames
$dimnames[[1]]
[1] "r1" "r2" "r3"

$dimnames[[2]]
[1] "c1" "c2" "c3"
```

List

Lists in R are different from vectors as they can contain different data types. For example, in the following example we have a number and two strings: